

The compose premises (R) , (L) , (Z)

Three small analytic lemmas for COMPOSE⁻, and the contract they actually close

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Abstract

Round 383 attacks the three named remainders of the reduced compose implication of [1]: an absolute Fejér/diagonal ratio (R) , an ℓ^1 bound (L) , and an edge scalar (Z) . What is proved: the Fejér identity and bilinear Cauchy–Schwarz; the exact $r298$ split $S_F = B(\omega, \omega) + B(\Delta, \omega + \beta)$; the exact L^2 identity $M_3 = ((\log m)^2/m^2) T_2 E_\pi(F^2)$; $\Lambda(n) \leq \log n$ by trial division; the H -rule $\text{pref} \leq \sqrt{m} + 1$; and the T1-floor algebra conditional on $g_i \geq 3/8$. What is *not* proved: a cofinal absolute R_0 , a cofinal $\gamma < 1/4$ from the von Mangoldt triangle, a family-uniform $E_\pi(F^2)$ bound, or a PNT-free finite-head bound on $|Z_{\text{loc}}|$. Exact period-2 fold pairing makes $B(\omega, \omega)$ of size $O(D/m)$, but the live window field has pair-residual median 0.512 — the emptiness of the window main term is a census, not a pairing theorem. On the 181-window surface one has $S_F \leq 4D$ and $L_1 \leq 3$ (measured maxima 2.917 and 2.126); $|Z_{\text{loc}}| \leq 4/5 < M$ on every FRAME-A and FRAME-B window, and fails on six χ windows. The T1-combo route does not carry (E_π spikes at kz111/117). Scramble seed 1 named-breaks (L) on the M_3 polylog. No RH claim.

Status. Lemmas 1–6 are proved (finite algebra). Lemma 3 is proved on exact period-2 sequences and *refuted* as a description of the live P_ω . Proposition 1 records (R) reduced to the census $R_0 = 4$. Proposition 2 records (L) reduced to $\Lambda = 3$ plus a measured slope, with the triangle and T1-combo routes closed honestly. Proposition 3 records (Z) reduced on FRAME-A ($Z_0 = 4/5$) and refuted family-uniformly on χ . No RH claim.

1 Recollection

Write $M = \sqrt{5/7}$, $H = \max(2, \lceil \sqrt{m} \rceil)$, $\text{pref} = (m + H - 1)/H$, $D = \sum_j P_j^2$, $L_1 = \sum_j |P_j|$, and $S_F = (1/H) \sum_t W_t^2$ for the Fejér energy of a real block field P of length m [1]. The reduced compose implication is

$$S_F \leq R_0 D \wedge L_1 \leq \Lambda \wedge |Z_{\text{loc}}| \leq Z_0 < M \wedge M_3 \leq \phi(m) \implies |Z| < M,$$

with $\phi(m) = (M - Z_0)^4 / (\text{pref}(m)^2 R_0^2 \Lambda^4)$ and $\text{pref}(m) \leq \sqrt{m} + 1$. If $L_1 \leq C m^\gamma$ replaces the absolute Λ , the same algebra demands $2\beta = 1 + 4\gamma$; T1-floor supplies $\beta = 1$, which beats this if and only if $\gamma < 1/4$ (at $\gamma = 1/4$ one needs $(\log m)^2$ bounded). Hypothesis H5 is a sister route, not an extra premise. Claim boundary: finite identities and named reductions. Not evidence for or against the Riemann hypothesis.

2 Shared machinery

Lemma 1 (Fejér identity and van der Corput). *For every real sequence P of length m and every $H \in \{1, \dots, m\}$, $|\sum P|^2 \leq \text{pref} S_F$ with $S_F = (1/H) \sum_t W_t^2 = \sum_{|h| < H} (1 - |h|/H) A(h) \geq 0$.*

The bilinear form $B(u, v) = (1/H) \sum_t W_t(u) W_t(v)$ is a Gram form: $B(u, v)^2 \leq B(u, u) B(v, v)$. Consequently $S_F \leq HD$.

The bound $S_F \leq HD$ is a theorem and is too coarse for composition (H grows like $\sqrt{m}$). Companion: exact period-2 witness $P = (1, -1)^8$, $H = 4$: $S_F = 1$, $D = 16$, $R = 1/16$; DC witness $P = 1^{16}$: $R = 3.6875 \leq H = 4$.

Lemma 2 (The *r298* split). *On the frozen positional Fejér kernel, $S_F = B(\omega, \omega) + B(\Delta, \omega + \beta)$ identically, with $\Delta = \beta - \omega$. The geometric edge of the canonical split satisfies $Z = Z_{\text{loc}} + t_{\text{bulk}}$ (Lean *canonical-split*); t_{bulk} is the bulk chain sum, not $\sum P_\Delta$.*

Lemma 3 (Exact period-2 fold). *If $P = (a, -a, \dots, a, -a)$ has even length and H is even, then every complete H -window sums to 0 and $S_F/D = O(1/m)$ (the residual is the $H - 1$ incomplete edge windows). In particular the window main term $B(\omega, \omega)$ would be $O(D_\omega/m)$ if P_ω were exactly period-2.*

Remark 1. Lemma 3 is *not* the live P_ω . On 181 windows the adjacent-pair residual $\sum |P_{\omega, 2k} + P_{\omega, 2k+1}| / \sum |P_{\omega, 2k}| + |P_{\omega, 2k+1}|$ has median 0.512. The measured emptiness $\text{med } B(\omega, \omega) / S_F = 0.0086$ is a low-pass census (window lag-1 correlation median -0.459), not a pairing identity. Fold-parity as a SATZ for the live field is therefore refuted; MAIN-empty remains a census.

Lemma 4 (L^2 identity). *Writing $q_i = |P_i|/L_1$, g_i the normalised local gap, $F_i = (mq_i / \log m) g_i$, $T_2 = \sum_i q_i / g_i^2$ and $\pi_i = (q_i / g_i^2) / T_2$, one has $M_3 = ((\log m)^2 / m^2) T_2 E_\pi(F^2)$ identically.*

Lemma 5 (Von Mangoldt triangle). $\Lambda(n) \leq \log n$ for every integer $n \geq 2$ (trial division: $\Lambda(p^k) = \log p$). *The triangle $L_1 \leq \sum_{\text{atoms}} |\Delta c_a|$ therefore costs at most one log per atom, not a power $\gamma < 1/4$.*

Lemma 6 (H -rule and ϕ). *At $H = \max(2, \lceil \sqrt{m} \rceil)$ one has $\text{pref} \leq \sqrt{m} + 1$. If $g_i \geq 3/8$ and $F_i \leq C_K$ then $M_3 \leq ((8/3)C_K(\log m)/m)^2$. Under $(R)(L)(Z)$ the sufficient M_3 -cap is $\phi(m) = (M - Z_0)^4 / (\text{pref}^2 R_0^2 \Lambda^4)$.*

3 Premise (R)

Proposition 1 ((R) , reduced). *On the 181-window surface (FRAME-A 89 + FRAME-B 8 + χ_3 42 + χ_4 42), $S_F \leq 4D$. The measured supremum is 2.916788 at FRAME-A *kz37* ($H = 15$). The constant $R_0 = 4$ is at most twice that supremum and strictly below the trivial envelope ($H_{\text{max}} = 29$). The mutant $R_0/2 = 2$ fails on 17 windows and on the DC toy. A cofinal absolute R_0 is not proved: Lemma 3 does not apply to the live P_ω , and $S_F \leq HD$ grows.*

Outcome: **REDUCED** to the census $R_0 = 4$, with the cofinal ratio a named remainder. χ is not an exception for (R) .

4 Premise (L)

Two routes were available.

T1-combo. Lemma 4 is a theorem (worst relative residual $1.4 \cdot 10^{-15}$ on 181 rows). Under the floor $g_i \geq 3/8$ one has $T_2 \leq (8/3)^2$, and indeed $T_2 \leq 1.329$ on the surface. The remaining factor is $E_\pi(F^2)$. It is *not* family-uniform polylog: $E_\pi = 52.276$ at FRAME-A *kz111* and $E_\pi = 75.691$ at FRAME-B *kz117* (the *r368* spikes). The T1-combo therefore does *not* kill (L) and T1 together.

Von Mangoldt / FAB. Lemma 5 gives $\Lambda \leq \log$, hence a triangle. The live total variation has log-log slope $+0.307$ on FRAME-A, which is *not* $< 1/4$. The triangle does not prove $\gamma < 1/4$.

Census. $L_1 \leq 3$ on all 181 windows (maximum 2.126 at FRAME-B kz124). The log-log slope of L_1 is +0.202 on the 42-rung ladder and +0.223 on FRAME-A — measured, strictly below $1/4$, not a theorem. Because the slope is positive, no cofinal absolute Λ follows from the census.

Proposition 2 ($((L)$, reduced). $L_1 \leq 3$ on the 181-window surface. The T1-combo is a theorem of the identity and a refutation of the E_π bound. The von Mangoldt triangle does not yield $\gamma < 1/4$. The mutant $\gamma = 1/4$ fails the T1-versus- ϕ algebra $((\log m)^2$ unbounded). Scramble seed 1 at $w = 9$ named-breaks the r306 polylog $M_3 \leq 1.069 (\log m)^2 / m^2$ ($2.288 \cdot 10^{-2} > 1.374 \cdot 10^{-2}$) while leaving $R_0 = 4$ and $\Lambda = 3$ intact — the named (L) break.

Outcome: **REDUCED**. Remaining: a cofinal $\gamma < 1/4$ or a family-uniform $E_\pi(F^2)$ bound.

5 Premise (Z)

$Z_{\text{loc}} = t_{\text{loc}} + \text{chain}$ is the $F = 0.20$ geometric edge of the canonical split (Lean `canonical_split` is the algebraic triangle, already proved). The geometric edge contains 167 to 4689 atoms on the surface — it is not a finite von Mangoldt head. A Chebyshev/Rosser bound $\psi(x) \leq 1.04x$ therefore does not supply a finite sum.

Proposition 3 ($((Z)$, reduced on FRAME-A, refuted on χ). On FRAME-A, $|Z_{\text{loc}}| \leq 4/5 < M$ holds on 89/89 windows (maximum 0.755675 at kz16). FRAME-B: $8/8$ (maximum 0.540). The 42-rung ladder: 42/42. On χ the bound $|Z_{\text{loc}}| < M$ fails on six windows: χ_3 kz{15, 19, 23, 33, 39} and χ_4 kz20 (maximum 1.021 at χ_3 kz15). Thus 175/181 windows satisfy $|Z_{\text{loc}}| < M$. A cofinal PNT-free bound is not proved.

Outcome: **REDUCED** on the canonical FRAME-A family with $Z_0 = 4/5$; **REFUTED** as a family-uniform statement on χ . Finite-head Chebyshev: **REFUTED**.

6 The contract

Taking the census constants $R_0 = 4$, $\Lambda = 3$, $Z_0 = 4/5$ inside the r378 implication, COMPOSE[−] is a theorem of those three inequalities plus $M_3 \leq \phi(m)$ on every FRAME-A window of the 181-surface. The three inequalities are census SATZ on that surface, not cofinal. χ blocks only (Z) . T1 remains open as a family-uniform C_K or E_π bound; the identity of Lemma 4 stands.

H5/PairMarginLaw is untouched (sister route). The r263 dictionary remains a named sorry. The 42-rung $q_N < 1$ certificate is already sealed and is not re-proved here.

7 Machine check

The companion script `verify_compose_premises.py` and the discovery probe `compose_premises_probe.py` gate the toys (period-2, DC mutant, Cauchy–Schwarz, L^2 identity, $\gamma = 1/4$ algebra, $\Lambda \leq \log$), the H -rule and ϕ , the $w = 9$ pins, the scramble M_3 -kill of (L) , kz37 (R -star), kz16 (Z -star), kz111 (E_π spike: T1-combo does not close as a theorem), and on `--full` the 181-row census including $Z = Z_{\text{loc}} + t_{\text{bulk}}$. Final line COMPOSE PREMISES VERIFIED. No RH claim.

References

- [1] S. Hamann, *The compose lemma, reduced*, `rh/problem/compose_lemma.pdf`, 2026.
- [2] TFPT verification module `v964_1star_coherence_census.py`, van der Corput S0.

- [3] Discovery probe `window_border_transfer_probe.py`, r_{298} split.
- [4] Discovery probe `weighted_l2_t1_probe.py`, L^2 identity.
- [5] `rh/lean/RH/PairBound.lean`, `canonical_split`.